

Imperial College London
Department of Earth Science and Engineering
MSc in Applied Computational Science and Engineering

Independent Research Project
Final Report

Solving the Wave Equation by Optimising a Discrete Loss

by

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repository: <https://github.com/ease-ada-lovelace-2025/irp-ld2022>

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Abstract

The numerical modelling of waves has broad applications across science and engineering. There are many ways to represent waves computationally, but each requires discretisation, relying on discretisation schemes like the finite difference method. Discretisation introduces errors that can compound over time, particularly when applied to the wave equation. Recently, the Optimising a Discrete Loss (ODIL) framework has been proposed as an efficient method for producing accurate approximations of the solution of Partial Differential Equations (PDE) by means of minimising a discrete loss function. ODIL does not require a time stepping approach, unlike classical numerical methods, which may offer accuracy and efficiency benefits when solving the wave equation. This work aims to explore the accuracy and performance characteristics of the ODIL framework when applied to the wave equation.

1 Introduction

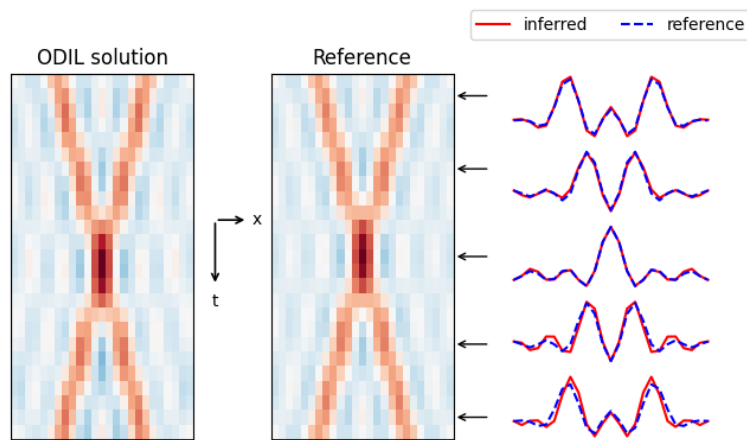


Figure 1: A test figure.

2 Methods

3 Results

4 Discussion

5 Conclusion

References